Ex.No.-1.2-Design and Draft the given 2D Sketches in modelling software.

AIM

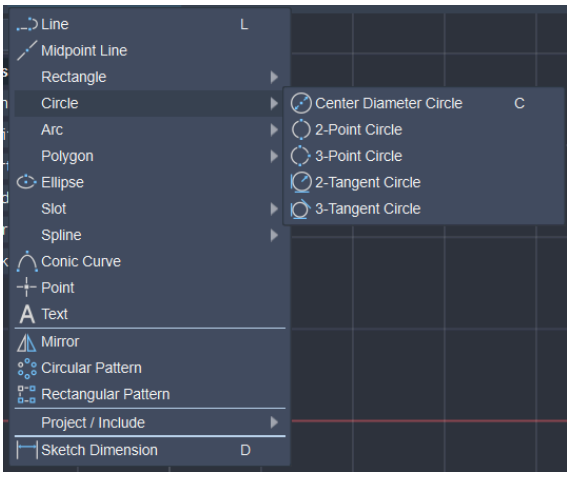
To sketch the given drawing with dimensions using fusion 360 tool.

REQUIREMENTS

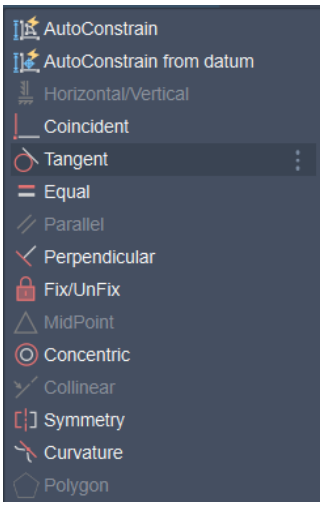
1. Autodesk fusion 360
2. Windows 10
3. 16 GB of RAM (integrated graphics recommend 1 GB or more)
4. 2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE

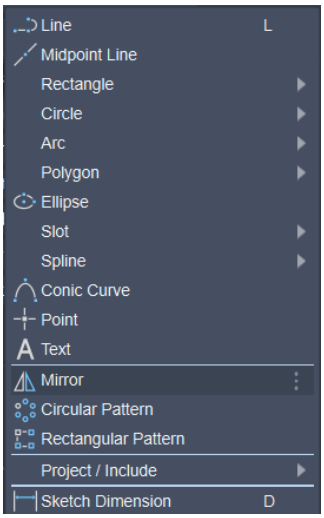
1. Open Autodesk Fusion and create a new design. Select **Create Sketch** from the Sketch section and choose the **Front plane** as the sketching plane.
2. After entering the sketch environment, select the **Line** tool from the Sketch toolbar. Draw the initial horizontal and vertical reference lines according to the given drawing and apply the required dimensions using the **Sketch Dimension** tool.
3. Create the main reference lines and locate the required centre positions according to the dimensions given in the question. Maintain a horizontal distance of **50 mm** between the required centre positions, and apply the vertical dimensions as specified in the drawing, including the **60 mm** and **20 mm** distances.
4. Use the **Circle** tool to create the required circular features. Draw the circles at the specified locations and apply their respective diameters as shown in the given drawing.
5. Construct the upper circular portion of the component by creating the required concentric circles. Apply the corresponding dimensions and use the centre points to maintain the required alignment.

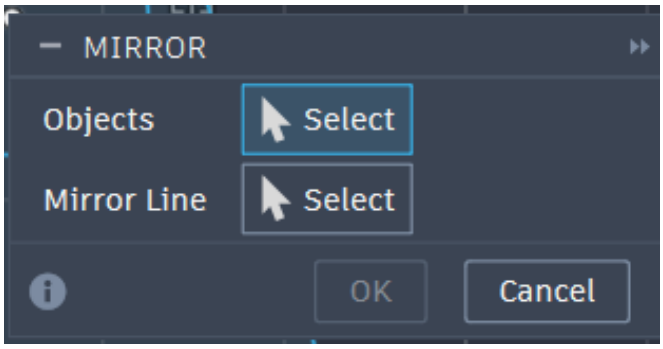


6. Next, create the outer profile of the component using the required **Line** and **Arc** tools. The curved portions are formed according to the dimensions given in the drawing, including the required **R32** and **R40** radii.
7. Apply the required geometric constraints and dimensions to position the curves correctly with respect to the previously created circles and reference geometry. This ensures that the sketch follows the proportions and measurements of the given component.

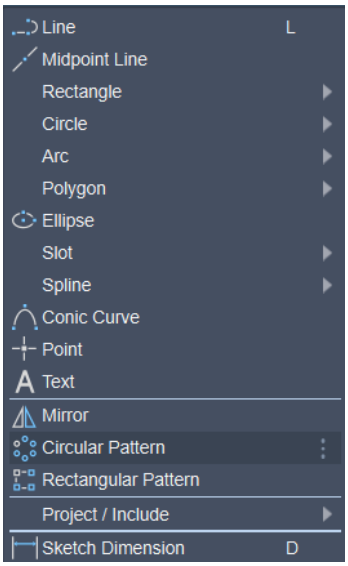


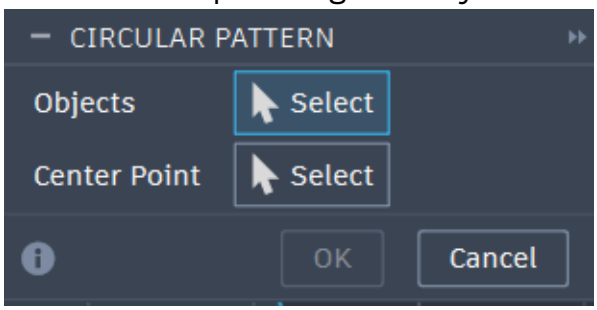
8. Create the two side circular features using the **Circle** tool. Position them according to the given horizontal and vertical dimensions and apply the required radius values, including **R15** and **R16**.
9. Create the lower circular feature at the centre of the bottom portion. Draw the required circles and apply the specified dimensions.
10. Construct the curved connections between the lower circular features and the main outer profile using the **Arc** tool. Apply the required radius values, including **R21** and **R20**, and adjust the curves so that they smoothly connect with the surrounding geometry.
11. After creating the basic geometry, use the **Mirror** option to reproduce the required symmetrical geometry about the appropriate centre/reference line. Select the geometry that has to be mirrored and choose the corresponding centre line as the mirror axis.



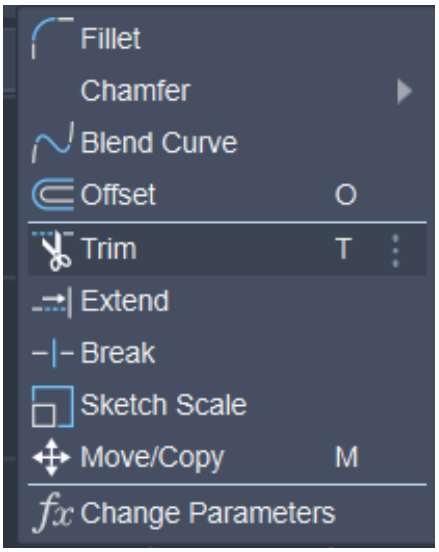


12. Once the mirrored geometry is created, check both sides of the sketch to ensure that the corresponding features are symmetrical and correctly positioned.
13. Create the central patterned feature using the required circles and arcs. Draw the central circle and construct the surrounding repeated curved portions, maintaining a spacing of **6 mm** between the arcs as shown in the drawing.
14. Apply the required dimensions to the central feature and use the appropriate pattern/mirror operation to obtain the repeated geometry with the correct arc spacing.



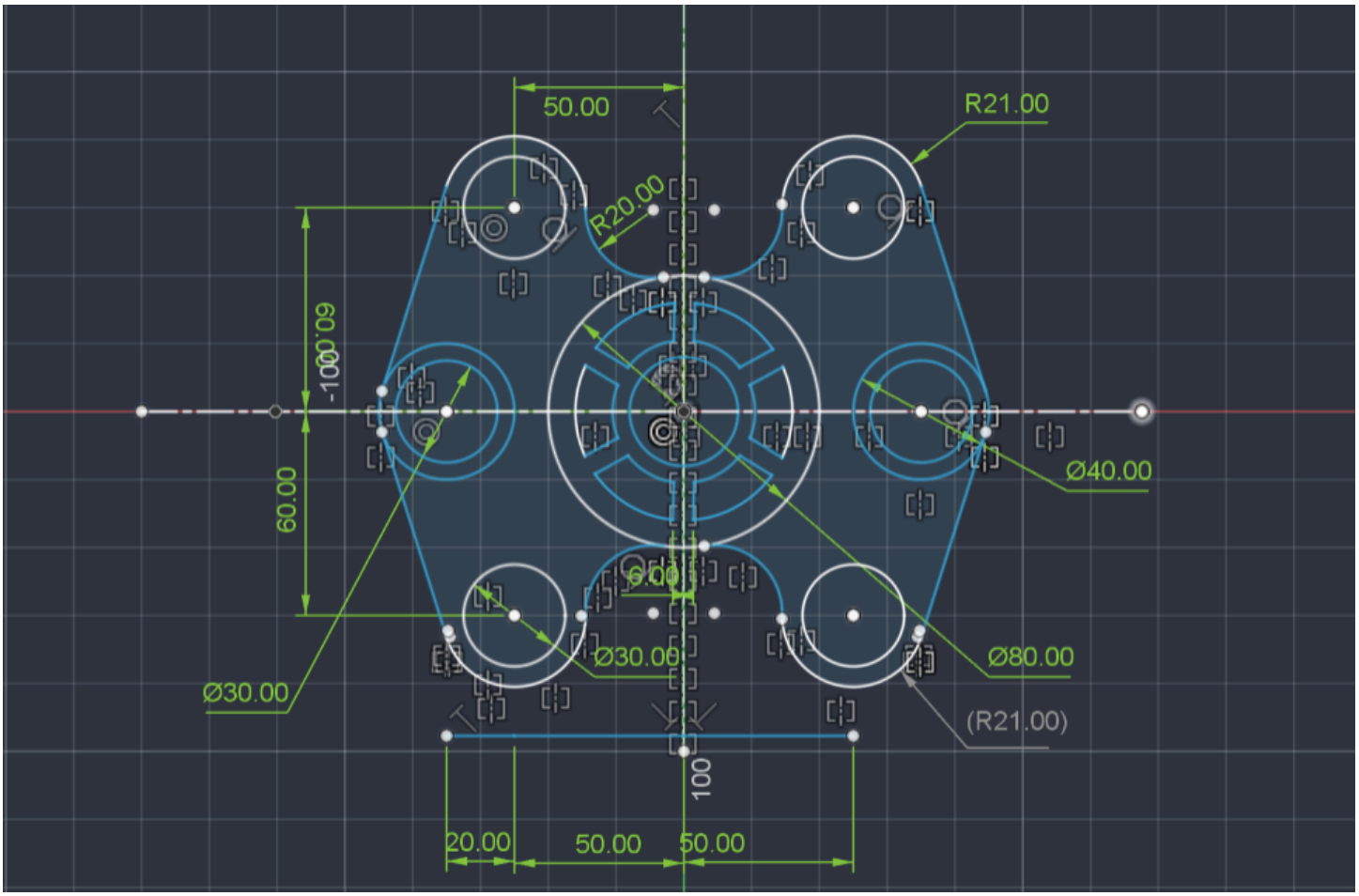


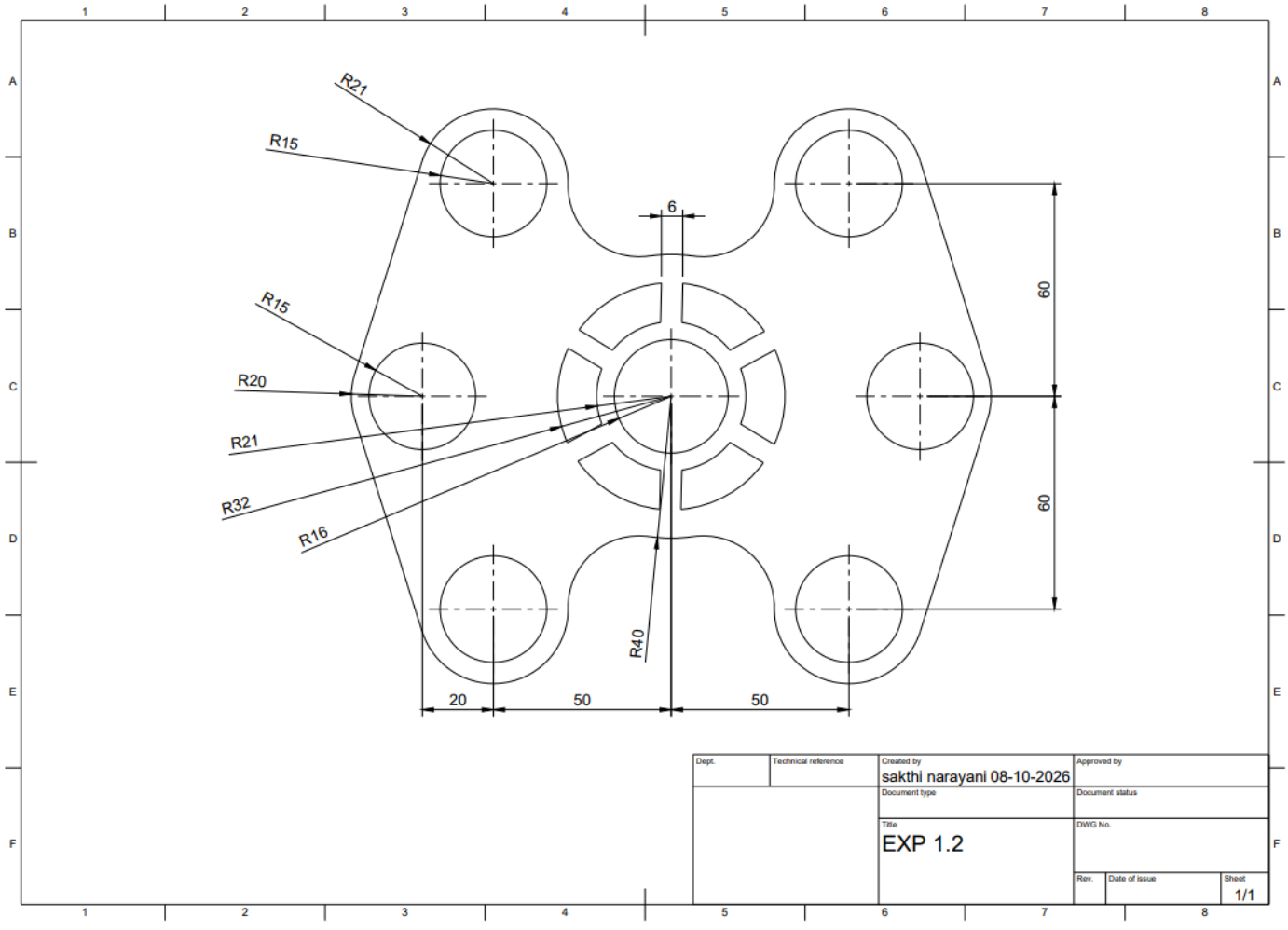
15. After completing the required geometry, use the **Trim** tool to remove the unwanted portions of circles, arcs and overlapping lines. Select the portions that are not part of the final profile and trim them carefully.



16. Continue trimming the unnecessary sections until the outer boundary and inner features match the given component drawing. Check that no unwanted construction or overlapping portions remain in the final sketch.
17. Verify all important dimensions and positions in the completed sketch. Check the **50 mm** horizontal dimension, the **60 mm** and **20 mm** vertical dimensions, the **6 mm** spacing between the central arcs, and the specified radii **R32, R40, R21, R20, R15** and **R16**.
18. Make the necessary corrections wherever required and ensure that the final geometry matches the reference drawing. The completed sketch should contain the upper circular portion, side holes, lower circular feature, outer curved profile and central patterned feature.
19. Finally, zoom out and display the complete drawing sheet containing the finished 2D sketch and its dimensions. Take a screenshot of the complete drawing for documentation.
20. Save the completed Fusion design. The final sketch is then ready to be included in the experiment documentation.

OUTPUT





Dept.	Technical reference	Created by	Approved by
		sakthi narayani 08-10-2026	
		Document type	Document status
		Title: EXP 1.2	DWG No.
			Rev. / Date of issue / Sheet: 1/1

RESULT

Thus the given sketch is drawn and drafted using fusion 360 tool.